

$$\Rightarrow \int (\sec n - \tan n)^2 \, dn.$$

sol

$$\text{Using } (a-b)^2 = (a^2 - 2ab + b^2)$$

$$a = \sec n \quad b = \tan n.$$

$$\int (\sec^2 n - 2 \sec n \tan n + \tan^2 n) \, dn.$$

$$\boxed{\tan^2 n = \sec^2 n - 1}$$

$$\int (\sec^2 n - 2 \sec n \tan n + (\sec^2 n - 1)) \, dn.$$

$$\int (\sec^2 n + \sec^2 n - 2 \sec n \tan n - 1) \, dn.$$

$$\int 2 \sec^2 n - 2 \sec n \tan n - 1 \, dn.$$

Now apply (give integration separately).

$$2 \int \sec^2 n \, dn - 2 \int \sec n \tan n \, dn - \int 1 \, dn.$$

$$\boxed{2 \tan n - 2 \sec n - n + C}$$